Department of Computer Science and Engineering

Program: BE

Date	23 June 2025	Maximum Marks	50
Course Code	**CS362AI**	Duration	90 min
6th Sem	VI Semester	CIE-II	

Network Programming and Security

	QUIZ	**M**	**L**	**CO**
1	If only the right half of the data is processed during each DES round, how does the left half contribute to the encryption?	2	L3	CO4
2	DES uses S-boxes in each round. What role do these S-boxes play in resisting cryptanalysis?	2	L3	CO4
3	What are the essential requirements a public-key cryptosystem must satisfy to ensure secure communication?	2	L3	CO5
4	In the Diffie-Hellman key exchange, if both users agree on a prime number p = 17 and a base g = 3, and User A chooses a private key a = 6, what public value does A send to B?	2	L3	CO5
5	During an HTTPS session, how do SSL/TLS protocols ensure that the client is communicating with the intended server and not an imposter?	2	L3	CO5
Sl. No.	**Test Questions**	**M**	**L**	**CO**
1a	Explain the working of the DES algorithm with a neat diagram. Discuss its strengths and weaknesses.	**10**	L1	CO4
2.a	What are the principles of public-key cryptosystems?	**05**	L3	CO4
2.b	Alice wants to send a confidential message to Bob using a public-key cryptosystem. Explain the steps she must follow to ensure that only Bob can read the message.	**05**	L4	CO4
3.a	What is public-key cryptanalysis and how can such systems be attacked?	**05**	L4	CO4
3.b	Bob receives a digitally signed message from Alice. How can he verify that the message is authentic and has not been tampered with?	**05**	L1	CO4
4	Explain the Diffie-Hellman key exchange algorithm with an example. Also, describe its key exchange protocol and how it is vulnerable to a man-in-the-middle attack.	**10**	L4	CO5
5	Explain how HTTPS ensures secure communication over the internet. Describe the role of SSL/TLS protocols in the HTTPS handshake process and mention key security services provided.	**10**	L3	CO5

Course Outcomes

CO1	Explore the variety of network programming concepts and protocols.
CO2	Analyse the interoperability of networking protocols and its usage.
CO3	Design the client/server communication on Unix platforms.
CO4	Investigate & Design the cryptographic algorithms to ensure secure transfer of secret keys and encryption/decryption of messages.
CO5	Demonstrate Network Programming and Cryptographic algorithms to solve real-world problems.

L1	L2	L3	L4	L5	L6	CO1	CO2	CO3	CO4	CO5	CO6
15	-	25	20	-	-	-	-	34	26	-	-